

ECON 8803: PS5

Rohit Borah & Mingyu Li

April 23, 2026

Consider the following random utility model for cars:

$$V_{ij} = \beta_q q_{ij} + \beta_p p_{ij} + \beta_c c_i + \varepsilon_{ij}, \quad j = 1, \dots, 6$$

where p_{ij} is price, q_{ij} is quality, and c_i is cost. The (fake) data for this problem set is available in CSV format is `ps5_data.csv`. The dataset includes choices and option characteristics for 1,000 individuals (`id`). The variable `choice` is an indicator variable equal to one for the car chosen by each individual. The variable `cs` enumerates the options available in each consumer's choice set (1, ..., 6). The data also includes three variables measuring the characteristics of 'rival' options available to the consumers:

- **dist_price**: Euclidean distance between price of product (i, j) and other products available, i.e., $\sqrt{\sum_{j' \neq j} (p_{ij} - p_{ij'})^2}$
- **dist_quality**: Euclidean distance in quality
- **dist_cost**: Euclidean distance in cost

1. Create a summary statistics table summarizing the characteristics of each car model chosen and not chosen across all consumers.

Summary Statistics Table: Choice by Car Model

Model	Chosen	n	Price	Quality	Cost	Dist_Price	Dist_Quality	Dist_Cost
1	Yes	61	10.11 (0.10)	1.07 (0.19)	1.47 (0.09)	28.76 (1.81)	6.79 (0.39)	1.24 (0.22)
	No	939	10.09 (0.11)	1.00 (0.20)	1.50 (0.10)	28.83 (1.85)	6.88 (0.41)	1.31 (0.22)
2	Yes	460	10.55 (0.30)	1.19 (0.20)	0.79 (0.10)	28.01 (1.94)	6.53 (0.42)	1.09 (0.19)
	No	540	10.56 (0.32)	1.20 (0.21)	0.81 (0.10)	28.12 (1.94)	6.53 (0.42)	1.05 (0.19)
3	Yes	236	11.85 (0.57)	2.00 (0.20)	0.97 (0.09)	26.23 (2.19)	5.31 (0.34)	0.85 (0.12)
	No	764	11.84 (0.54)	2.00 (0.20)	1.01 (0.10)	26.06 (1.98)	5.30 (0.33)	0.85 (0.11)
4	Yes	187	20.42 (1.15)	3.95 (0.18)	0.67 (0.09)	22.01 (1.48)	4.99 (0.27)	1.30 (0.20)
	No	813	20.73 (1.39)	4.01 (0.20)	0.70 (0.10)	22.20 (1.65)	5.04 (0.29)	1.23 (0.21)
5	Yes	23	20.67 (1.34)	4.43 (0.18)	1.50 (0.09)	22.31 (1.78)	5.59 (0.33)	1.27 (0.19)
	No	977	21.31 (1.36)	4.50 (0.20)	1.50 (0.10)	22.64 (1.72)	5.67 (0.36)	1.30 (0.21)
6	Yes	33	34.35 (2.27)	6.00 (0.22)	0.98 (0.09)	45.04 (4.96)	8.37 (0.50)	0.86 (0.12)
	No	967	34.29 (2.03)	5.99 (0.20)	1.00 (0.10)	44.84 (4.48)	8.38 (0.47)	0.85 (0.12)

2. Estimate a conditional logit model for the random utility model in Equation (1).
 - It does make sense that consumers prefer lower prices and lower cost cars. While I was initially surprised that quality also had a negative coefficient, it does make sense given that consumers prefer lower price cars. To a certain extent with cars, you get what you pay for, and that is reflected here by consumers choosing both lower price and therefore lower quality cars. This is an issue of collinearity within price and quality.
3. Now add the distance variables in the above model.
 - By accounting for Euclidean distance, the coefficient for quality turns positive due to better contextualization of cars with better quality within a given price band. The quality coefficient changes signs from negative to positive because adding the distance variables helps to soak up some of the variance between models.

	<i>Dependent variable:</i>	
	choice	
	Model 2	Model 3
price	−0.092*** (0.025)	−0.235*** (0.072)
quality	−0.048 (0.101)	0.538* (0.309)
cost	−2.744*** (0.140)	−2.988*** (0.174)
dist_price		0.053** (0.024)
dist_quality		0.029 (0.145)
dist_cost		0.087 (0.258)
Observations	1,000	1,000
Log Likelihood	−1,355.877	−1,350.184
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

4. Next, use the distance variables to test the null hypothesis of IIA preferences. In particular, for the conditional logit model you estimated in part 3, test the null hypothesis that variables measuring characteristics of rival products are excluded (as in Equation 1). What do you observe? (Note: Some of the distance variable are highly correlated with each other, so you may conduct this test by adding one distance variable at a time.)

Hypothesis Testing IIA Preferences				
Variables Added	df	LogLik	χ^2	p-value
dist_price	1	-1350.298	11.157	0.0008
dist_quality	1	-1354.041	3.673	0.0553
dist_cost	1	-1353.991	3.771	0.0521
dist_price + dist_quality	2	-1350.242	11.271	0.0036
dist_price + dist_cost	1	-1350.205	11.157	0.0034
dist_quality + dist_cost	2	-1352.516	6.722	0.0347
all three	3	-1350.184	11.385	0.0098

- Based on the regression analysis, dist_price is the most important factor determining the likelihood that a car is purchased. Because the p-value for all three is 0.0098, it is most likely that IIA is violated.
5. The table below shows the average elasticity of substitution between products obtained from a mixed-logit model for the utility model in Equation (1). The columns (1 – 6) report the elasticity of demand for product i (rows) with respect to the price of product j (columns). The second and third column summarizes the average market-share (demand) and price.

Model	Share	Price	1	2	3	4	5	6
1	0.06	10.09	-3.29	1.89	1.31	0.55	0.25	0.06
2	0.46	10.56	0.17	-1.61	1.15	0.65	0.12	0.06
3	0.24	11.85	0.20	1.96	-2.96	0.67	0.16	0.08
4	0.19	20.67	0.09	1.17	0.70	-4.05	0.16	0.28
5	0.02	21.30	0.20	1.05	0.84	0.78	-4.51	0.23
6	0.03	34.29	0.02	0.26	0.19	0.62	0.10	-3.11

- (a) Based on the demand elasticities shown above, what can you say about the effect of increasing price of the most popular product (2 lines max)?
- Increasing the price of Model 2 increases the demand for all other models and decreases demand for itself.

- (b) Now, for the conditional logit model you estimated in part 2, compute the elasticity matrix and show it in the same format as the one shown above.

Conditional Logit Model: Elasticity Matrix								
Model	Share	Price	1	2	3	4	5	6
1	0.07	10.090	-0.865	0.438	0.246	0.395	0.042	0.075
2	0.45	10.560	0.066	-0.535	0.246	0.395	0.042	0.075
3	0.23	11.850	0.066	0.438	-0.846	0.395	0.042	0.075
4	0.21	20.670	0.066	0.438	0.246	-1.511	0.042	0.075
5	0.02	21.300	0.066	0.438	0.246	0.395	-1.922	0.075
6	0.02	34.290	0.066	0.438	0.246	0.395	0.042	-3.086

- (c) Intuitively (without using any math) what can you say about the IIA property across mixed logit vs conditional logit in this context (3-4 lines max)?
- In the conditional logit model, cross-price elasticities are identical because IIA constrains the substitution effect. Mixed logit captures substitutes and elasticities, showing us that Models 1 and 3 are substitute goods for 2 while 6 is less relevant.